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被街道重塑的大脑

The Brain That Learned London



熟记伦敦街道，会怎样改变大脑？

英文原文

Name two addresses and a London **cabbie** will **rattle off** the **quickest route**. That **prowess** is **hard-won**. To earn a **licence**, **aspiring** drivers must **conquer** "the Knowledge", an **arduous examination** on about 25,000 streets within six miles of Charing Cross. **Hopefuls typically** spend three to four years **committing** this **labyrinth** to memory, and only about half **qualify**.

A study published in PNAS in April 2000 **scanned** 16 male cabbies and 50 other men. The **hippocampus** is a seahorse-shaped structure **vital** to memory and **navigation**. In the cabbies, its **rear**, or **posterior**, **portion** was larger than in the other men, and its front, or **anterior**, portion smaller. Longer service went with a larger rear portion.

Yet the link left the cause **uncertain**. Perhaps an **innate flair** for **spatial** tasks **lured** people to the **occupation**, or perhaps **stressful** driving was the **culprit**. A December 2006 study **dispelled** the second **notion**: cabbies had more rear **grey matter**, the tissue **dense** with **nerve** cells, than bus drivers, **confined** to **predetermined** routes, of **equivalent** experience and stress.

A December 2011 **longitudinal** study **tackled** the first, scanning 79 **trainees** and 31 non-trainees at **enrolment** and three to four years later. Their brains began **alike**. Only the 39 who qualified **exhibited** an **expansion** of rear grey matter, so the **sequence** suggests learning **preceded** the **enlargement**.

The gain appeared to **entail** a **trade-off**: qualified drivers **trailed** the non-trainees on one test of recalling an **intricate** figure. The **cohorts** were small and **exclusively** male, yet the studies **imply** the healthy adult brain can **remodel** itself through **intensive** learning. Every street learned, it seems, **occupied** room once **claimed** by something else.

中文译文

随便说出两个地址，伦敦的出租车司机就能脱口说出最快的路线。这份本领来之不易。要拿到执照，有志向的司机必须攻克“知识大考”，这是一场艰巨的考试，考的是约25,000条街道，都在查令十字周边6英里以内。考生们通常要花三到四年把这座迷宫记在脑子里，并且只有大约一半，最终会合格。

一项研究发表于《美国国家科学院院刊》（2000年4月），扫描了16名男性出租车司机和50名其他男性（的大脑）。海马体是一个形似海马的结构，对记忆和导航都是至关重要的。在出租车司机身上，海马体靠后的（也就是后面的）部分比其他男性的更大，而靠前的（也就是前面的）部分则较小。从业时间越长，后部也越大。

然而，这种关联让原因仍然是不确定的。也许是一种天生的、擅长空间任务的天赋把人们吸引到了这个职业里，又或许紧张的驾驶才是罪魁祸首。2006年12月的一项研究打消了第二种想法：出租车司机后部的灰质更多，灰质即密布神经细胞的组织，比公交司机的多；这些公交司机局限于预先确定的线路，经验和压力（与出租车司机）都是相当的。

2011年12月的一项纵向的研究（着手）处理第一种想法（即天赋把人吸引到这一行），扫描了79名学员和31名非学员，分别在入学时和三到四年后（各扫描一次）。他们的大脑一开始都一样。只有最终合格的39人表现出后部灰质的扩大，因此这一先后顺序表明，学习先于（灰质的）增大发生。

这种收获似乎必然带来某种代价：合格的司机落后于非学员，是在“回忆一幅复杂的图形”这项测试上。这些研究的样本群体规模小，而且专门都是男性，但这些研究暗示/意味着，健康的成人大脑可以重塑自身，靠的是高强度的学习。看来，每记住一条街道，都占据了一块空间，这块空间原本被别的东西占用着。

词汇与表达

cabbie n.
出租车司机
addresses and a London cabbie will rattle off the

rattle off phr.
脱口说出

quickest route n.
最快的路线
will rattle off the quickest route.

prowess n.
本领
That prowess is hard-won.

hard-won adj.
来之不易

licence n.
执照

aspiring adj.
有志向的

conquer v.
攻克

arduous adj.
艰巨的
an arduous examination

examination n.
考试

Hopefuls n.
考生们

typically adv.
通常

committing v.
记
committing this labyrinth to memory

labyrinth n.
迷宫

qualify v.
合格
and only about half qualify.

scanned v.
扫描了

hippocampus n.
海马体
The hippocampus

vital adj.
至关重要的

navigation n.
导航
vital to memory and navigation.

rear adj.
靠后的
the cabbies, its rear, or posterior, portion

词汇与表达（续）

posterior adj.
后面的
its rear, or posterior, portion

portion n.
部分

anterior adj.
前面的
its front, or anterior, portion

uncertain adj.
不确定的
Yet the link left the cause uncertain.

innate adj.
天生的
an innate flair for spatial tasks

flair n.
天赋

spatial adj.
空间
spatial tasks

lured v.
吸引

occupation n.
职业

stressful adj.
紧张的

culprit n.
罪魁祸首

dispelled v.
打消了
study dispelled the second notion:

notion n.
想法

grey matter n.
灰质
cabbies had more rear grey matter, the tissue dense with

dense adj.
密布

nerve n.
神经

confined adj.
局限于

predetermined adj.
预先确定的
confined to predetermined routes

equivalent adj.
相当的
of equivalent experience and stress.

longitudinal adj.
纵向的
longitudinal study

词汇与表达（续）

tackled v.
处理

trainees n.
学员

enrolment n.
入学

alike adj.
一样
Their brains began alike.

exhibited v.
表现出

expansion n.
扩大
an expansion of rear grey matter

sequence n.
先后顺序

preceded v.
先于
learning preceded the enlargement.

enlargement n.
增大

entail v.
必然带来
to entail a trade-off:

trade-off n.
代价
appeared to entail a trade-off

trailed v.
落后于

intricate adj.
复杂的

cohorts n.
样本群体
The cohorts were small

exclusively adv.
专门

imply v.
暗示/意味着
yet the studies imply

remodel v.
重塑
can remodel itself

intensive adj.
高强度的

occupied v.
占据了

claimed v.
占用

文章简述

- 训练：伦敦出租车司机通常花三四年记住约2.5万条街道，只有约一半学员通过执照考试。
- 差异：海马体是参与记忆和认路的脑结构；出租车司机的海马体后部比其他男性大、前部小，但最初不知道这是天生差异还是工作造成的。
- 排查：公交司机也要长期驾驶，却只走固定路线；与经验和压力相近的公交司机比较，出租车司机后部富含神经细胞的灰质仍更多。
- 追踪：学员与未参训者起初脑部相似，三四年后只有考过者的后部灰质增加，说明学习发生在大脑变化之前，支持训练可能改变成人大脑。
- 取舍：样本小且全为男性，考过者在一项复杂图形记忆测试中还落后于未参训者；认路能力增强，未必等于各种记忆都增强。